

INSTRUMENT DESIGN TIER

EXECUTIVE CREDENTIAL SYNTHESIS

AARON RAY

AI Emotional Benchmarking • Forensic Audiovisual Analysis • Artificial Senses Systems

Target Package Architecture

Navy • Cream • IBM Plex Mono • Bebas Neue Headers • Gold Rules

Prepared for strategic positioning in advanced AI sensory & emotional intelligence benchmark design.

01

ENTERTAINMENT INDUSTRY CREDENTIAL BASE

Operational Scale Statement

Over 35 years directing human emotional response at scale across a major portfolio of commercial media properties, managing packaging, production, and executive execution pipelines that generated substantial global audience engagement and commercial performance.

Core Creative Elements Analyzed

Continuous analysis of audience behavioral engagement metrics, tracking the interplay of soundtrack structures, comedic timing, and dramatic narrative architectures to maximize viewer completion rates and retention.

This foundation in mass-scale emotional direction provides the empirical base layer for constructing valid, predictive emotional benchmarks in AI systems.

02

FORENSIC AUDIOVISUAL EXPERTISE

Forensic Phonetics & Witness Qualifications

Direct alignment with the foundational principles established in Hollien (1990), establishing core competencies in forensic acoustics, voice identification protocols, and the scientific authentication of audio evidence within legal chains of custody.

Acoustic Signal Analysis & Trained Listener Frameworks

System integration of Schiller & Koster (1998) methodologies, standardizing trained-listener verification loops to isolate micro-auditory variations, acoustic anomalies, and speech pattern anomalies.

Psychological Stress Tracking & Voice Security

Application of Acoustics of Crime parameters to track physiological stress signatures, frequency fluctuations, and sub-vocal indicators within voice streams. These parameters serve as the direct foundation for developing modern synthetic voice identification engines and deepfake detection software.

03

AI/ML TECHNICAL CREDENTIALS

Frontier Architectural Frameworks

Direct technical proficiency across machine learning models and frameworks, encompassing early implementations in TensorFlow and YOLOv3 object detection environments, alongside deep data tuning inside advanced natural language processing architectures.

Generative Music & Audio Networks

Technical execution tracking across advanced neural audio generation pipelines, including Google's Magenta platform, OpenAI Jukebox architectures, and digital acoustic sequence engines like EarSketch.

Specialized Deep Learning Credentials

Academic and practical training certifications in deep learning, sequence modeling, and natural language processing architectures from leading research institutions, including Johns Hopkins University and the University of Michigan.

04

THE ARTIFICIAL SENSES FRAMEWORK

Core Thesis

Artificial Senses is the use of technology to augment, measure, and regulate human emotional and perceptual response—across therapeutic, legal, and entertainment domains.

PILLAR I

MUSIC AS MEDICINE

Engineering non-pharmacological behavioral regulation loops using targeted acoustic patterns and structured audio environments to manage attention and sensory processing, serving as a non-chemical intervention model for pediatric ADHD populations.

PILLAR III

FORENSIC AUTHENTICATION

Building robust, automated verification workflows that leverage multi-modal models to flag synthetic media generation, verify metadata, and preserve evidentiary integrity.

PILLAR II

JUDICIAL ANALYTICS

Processing and mapping public behavioral signals, structural micro-expressions, and linguistic patterns from legal decision-makers to decode cognitive biases and predict systemic litigation outcomes.

PILLAR IV

NARRATIVE PERSUASION

Calibrating the alignment of imagery, soundscapes, pacing, and dialogue setups against specific psychographic profiles to maximize narrative impact and retention.

Aaron Ray is uniquely qualified to design, validate, and publish advanced AI emotional and sensory benchmarks based on a rare convergence of cross-domain disciplines:

- 1 **Decades of Practical Mastery**
Extensive professional experience directing human emotional response at scale within commercial entertainment environments.
- 2 **Academic & Forensic Grounding**
Structural knowledge base spanning forensic phonetics, voice print identification, and real-world acoustic surveillance analysis.
- 3 **Hands-On Machine Learning Expertise**
Deep technical proficiency building, training, and fine-tuning neural networks, audio generation systems, and agentic CLI pipelines.
- 4 **Advanced Multi-Modal Integration**
Proven deployment of top-tier brain encoding foundation models (such as Meta's TRIBE v2 framework) to map digital media assets directly to biological neuro-imaging data.
- 5 **Published Benchmark Track Record**
Demonstrated capability in compiling rigorous, objective metrics, building on the methodology established in the previously published Kingsfield Legal Citation Benchmark to eliminate systemic evaluation bias.